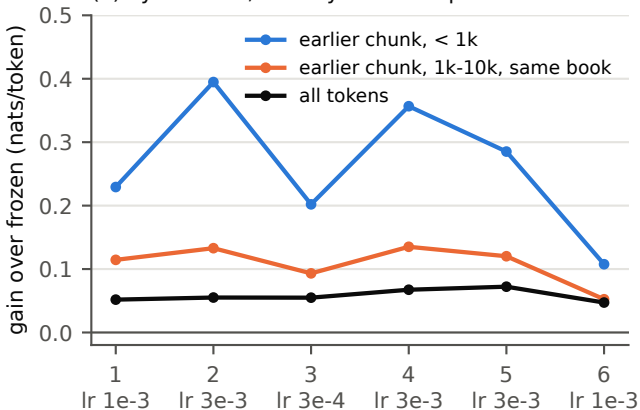
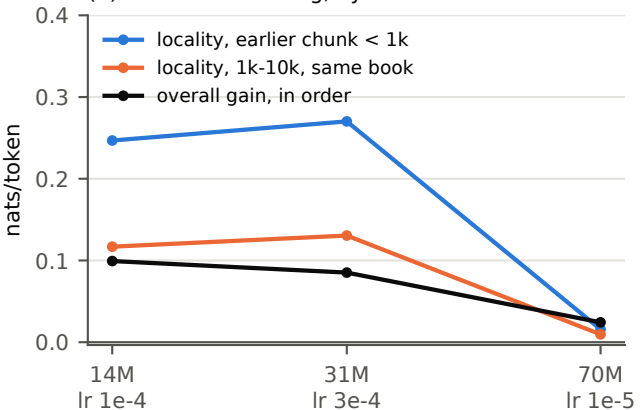

(a) Pythia-14M, one layer's MLP updated
gain over frozen (nats/token)
0.5
0.4
0.3
0.2
0.1
0.0
earlier chunk, < 1k
earlier chunk, 1k-10k, same book
all tokens
1
lr 1e-3
2
lr 3e-3
3
lr 3e-4
4
lr 3e-3
5
lr 3e-3
6
lr 1e-3


(b) full online learning, by model size
nats/token
0.4
0.3
0.2
0.1
0.0
locality, earlier chunk < 1k
locality, 1k-10k, same book
overall gain, in order
14M
lr 1e-4
31M
lr 3e-4
70M
lr 1e-5